

Guidance document for non-invasive determination of high voltage of X-ray tubes

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1 Introduction

An X-ray tube produces X-rays by emitting the electrons from the cathode and accelerating the electrons between the cathode and the anode. When the electrons hit the anode, they rapidly lose their energy either by ionization, mostly generating heat, but also characteristic X-rays, or by radiative processes, producing bremsstrahlung. The X-ray spectrum is composed of the latter two. The X-ray photon energy is equal or smaller than the energy that the electron loses in the interaction, which can be anywhere between 0 and the electron kinetic energy. Considering that the electron kinetic energy at the anode is equal to the product of the electron charge and potential between the cathode and the anode, the maximum photon energy (spectrum end-point) is directly determined by the potential difference between the cathode and anode.

The tube voltage does not only determine the spectrum end-point, but has major influence on the whole photon spectrum, alongside other parameters such as anode material and angle, inherent and added filtration, and the distance from the tube. The X-ray photon spectrum determines the penetration capability of photons in materials, the contrast of X-ray images, and also more specifically half-value layer, response of radiation detectors and dosimeters (all detector responses are dependent on photon energy, at least to some extent), values of conversion coefficients between different quantities etc. All these have an impact on practical radiation protection. Even though spectrometry provides the most comprehensive information about the X-ray spectrum, in the past it has been complicated to perform and often considered impractical. New simplified spectrometry procedures and other methodologies used to estimate quantities influencing the spectrum, such as the tube voltage, are described in this document

The tube voltage can be determined by invasive or non-invasive methods. Invasive methods use a suitable meter or a high-voltage divider connected directly to the high-voltage generator of the X-ray tube. Non-invasive methods determine the tube voltage without physically contacting or altering the X-ray tube or its circuitry and rely on observing external signals. The common approaches include X-ray spectrometry, kVp meters and X-ray multimeters (XMMs).

The Bureau International des Poids et Mesures (BIPM) Consultative Committee for Ionizing Radiation (CCRI) recognized the importance of tube voltage measurements for dosimetry, and it included the tube voltage in the Calibration and Measurement Capabilities (CMC) service categories list (more specifically, in CCRI Section I: X- and gamma rays, charged particles). Currently, only two countries have CMCs published for tube voltage: Greece, using an invasive high voltage divider and Netherlands, using an HPGe detector calibrated with spectral lines from known radionuclides (BIPM, 2026a). Considering that these CMCs go through rigorous peer review, and high standards of proof of competence and technical capabilities are required, it can be considered that both of these methods can be used at the highest metrological level for tube voltage measurements.

This document is dedicated to the non-invasive tube voltage measurements needed to establish reference X-ray radiation fields used in calibration of radiation protection dosimeters according to ISO 4037 (ISO, 2019). However, tube voltage measurements are also necessary when establishing reference fields used for calibration and testing of dosimeters and XMMs for quality control in diagnostic radiology (IEC, 2023; IEC, 2024; IEC 2025; IAEA, 2007), reference fields for calibration of radiotherapy chambers (IAEA, 2009; IAEA, 2024), and for other applications. The methods and conclusions described in this document are therefore not limited to calibration of radiation protection dosimeters.

The methods presented here complement a simple spectrometric method described in an ISO standard ISO 16526-3 (ISO, 2011b). There, the tube voltage in kV corresponds to the maximum energy of the X-ray spectrum (expressed in keV), and is equivalent to that point where the decaying slope of the energy spectrum meets with the abscissa.

2 High voltage generators, voltage dividers and motivation for this document

In ISO 4037-1, it is currently considered that only invasive measurements using calibrated resistor chain (voltage divider) are accurate enough to realize all X-ray reference fields, and ISO 4037-1 states that using spectrometry is not recommended. Voltage dividers are currently considered the gold standard for tube voltage measurements when they are used according to ISO 16526-1.

However, at the moment of writing of this document (2026), many calibration laboratories either did not yet procure voltage dividers, or they do not use them in regular work. In a questionnaire organized by the 22NRM07 GuideRadPROS project in 2023, only 5 out of 40 respondents replied that X-ray tube high voltage measurements are adequately covered in ISO 4037, and many respondents reported problems with using voltage dividers. Many respondents also suggested that other methods should be explored and potentially included in the future revisions of the ISO 4037 series.

2.1 Basic considerations about high voltage generators and voltage dividers

An example simplified diagram a voltage divider is illustrated in Figure 2.1.

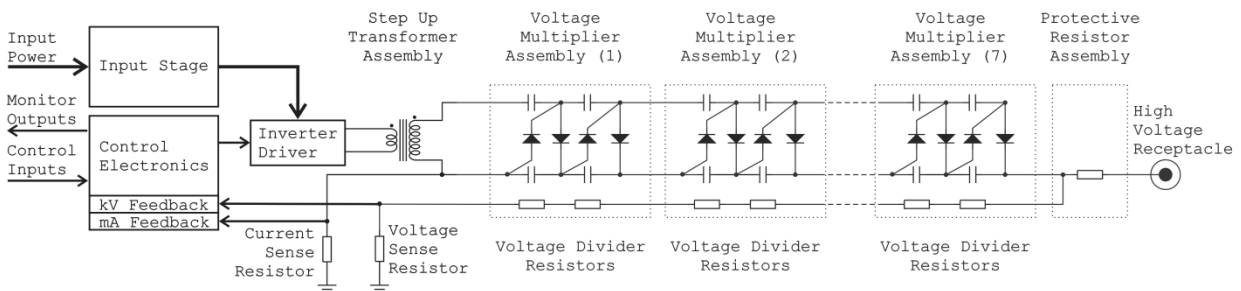


Figure 2.1 Simplified diagram of a high voltage generator tank

Input mains voltage is rectified at the input stage and filtered. This almost DC voltage is then used to power the inverter to drive the step up transformer at high frequency (e.g. 25 kHz to 45 kHz). Step up transformer is then powering several stages of voltage multipliers consisting of diodes and capacitors so the desired high voltage can be produced, of the order of 200 kV per generator in unipolar systems. As a part of voltage multiplier assembly, there are also voltage divider resistors in a high voltage tank filled with oil for electrical insulation. At the output, there might be a protective resistor (series output limiting resistor), which, in the eventual case of output short circuit or X-ray tube arcing, helps dissipate the power, protecting the generator from overload. Intermittent arcing will not damage the power supply unless there is a repetitive or continuous arcing in which case the protective resistor power handling rating could be exceeded, causing it to thermally overheat and fail. At the very output of the multiplier, before the protective resistor, there is a voltage divider resistor chain connected to the output voltage measurement and voltage regulator feedback control loop. At one side of the output of the transformer, there is a current sense resistor for monitoring and regulating X-ray tube current as a part of a current control feedback loop.

Modern X-ray high voltage generators have an internal voltage divider resistor chain connected to the generator output for a voltage regulation feedback loop, which allows the user to set and monitor high voltage at the output. As an example, accuracy for a typical modern generator is $\pm 1\%$, with reproducibility better than ± 0.05 kV. Ripple, defined as the difference between the maximum and the minimum voltage divided by the maximum voltage, is less than 5 V/mA (at 225 kV) to 10 V/mA for bipolar higher voltage models (320 kV and 450 kV). It can be concluded that at 20 mA, the ripple is in the worst case 200 V, (0.7 % at 28 kV, 0.13 % at 150 kV), which is less than 1 %, usually. This ripple is very

low in comparison to old designs that used 12-pulse full-wave rectification with the ripple between 3 % and 4 % of the peak voltage.

3 ISO 4037 reference fields and requirements for tube voltage

Calibration of radiation protection dosimeters is in most cases performed in reference fields realized according to ISO 4037-1 (ISO, 2019). These fields are mostly based on the measurements of air kerma and subsequent application of conversion coefficients from air kerma to operational quantities. X-ray reference fields can be either matched or characterized. Matched fields are realized according to the strict criteria for different parameters that affect the X-ray beam spectrum. These parameters are given in the standard, and with appropriate validation, it is allowed to use standard values of conversion coefficients given in ISO 4037-3. Requirements for characterized fields are less strict, but the laboratory needs to characterize the fields, either by using spectrometry and calculating the corresponding conversion coefficients, or by measuring operational quantities directly, using an appropriate secondary standard.

ISO 4037-1 gives the requirements for both characterized and matched reference fields. The requirements for all parameters are given having in mind the overall uncertainty goal for calibration in terms of operational quantities (6 % to 10 %, $k = 2$), as well as the uncertainty goal for conversion coefficients of 4 % ($k = 2$). There are several requirements related to the tube voltage, considering that it has major influence on the conversion coefficient: deviation of mean value of the tube potential during irradiation from nominal value, uncertainty of the tube potential, stability of the tube potential, and the ripple of the generating potential. There is also a requirement for the protective resistor in tube housing: it should either be absent, or it should be taken into account by correcting the tube voltage due to the current dependent voltage drop over the resistor. ISO 4037-1 gives procedure for dealing with protective resistor.

The requirements for the tube voltage depend on radiation quality, angle of incidence and operational quantity, and the limits are the same for deviation from nominal value, uncertainty of the tube potential (with $k = 2$) and stability of the tube potential (the limits for the ripple are twice as large as the limits given for the deviation from the nominal value). Uncertainty limit for all quantities at 0° , for all radiation qualities above L-55, N-60, W-60 and H-80 is 5 %. For radiation qualities L-35, N-25, N-30, N-40, W-30, W-40, H-40 and H-60, the limit is 5 % for operational quantities at 0.07 mm depth, 3 % at 3 mm depth and 1.5 % at 10 mm depth. Limits for other cases are given in ISO 4037-1 table 7 (ISO, 2019). In general, the requirements are stricter for lower voltage and higher irradiation angles.

The most important question that is dealt with in this document is whether the non-invasive measurement methods can fulfill these uncertainty requirements.

4 Voltage definition and measurement problems

ISO 4037-1 defines tube potential as the potential difference between cathode and anode of the X-ray tube, and it defines generating potential as the potential difference between the positive and negative output of the high voltage generator (ISO, 2019). The latter can generally be measured by a voltage divider. However, the same high voltage output may result in different tube voltages (and therefore different X-ray spectra) using different X-ray tubes, depending on the electrical circuits (e.g. due protective resistors, cable resistance, etc.).

The maximum ripple allowed according to the ISO 4037-1 is 10 % for radiation qualities above N-60, and the limit decreases with decreasing voltage (ISO, 2019). Ripple means that the generator output changes during the exposure, and the tube voltage also changes, causing the spectrum and all other related quantities to change. On the other hand, there is a strong need to describe the high voltage using a single number. Different quantities can be defined, such as maximum peak voltage (kVp) – maximum value of high voltage during an exposure, average peak voltage, average voltage, practical peak voltage (PPV) – weighted average voltage, according to the definition in IEC 61676 (IEC, 2023) etc. Another issue with the high voltage value is at which point during the exposure the voltage measurement is started and stopped, i.e. are the pulse rise and decay included, is the first peak included etc. ISO 4037-1 does not discuss the definition of the tube voltage quantity, however the ISO 16526-1 standard states that the method it describes is for measurement of average high voltage of constant potential (DC) X-ray systems on the secondary side of the high voltage generator (ISO, 2011a). In case of the constant potential generator, the values of all high voltage quantities would be equal (excluding pulse rise and decay).

As the ripple increases, the difference between the quantities also increases. Currently, high-voltage generators based on high frequency switching technologies with ripple below 0.1 % are commercially available (ISO, 2019). An example is a high-voltage generator with ripple of 5 V/ mA, where the usual range of currents is between 1 mA and 30 mA. This causes ripple below 0.1 % for most uses. In such cases, tube voltage can be considered constant, and any deviations can be considered in the uncertainty. Otherwise, the laboratory should consider high-voltage ripple in accordance with the uncertainty goals.

5 X-ray spectrometry

5.1 Equipment and basics

Spectrometric determination of tube voltage is relatively easily accessible method, especially if the laboratory already implemented in-beam X-ray spectrometry. The determination of the spectrum end-point is independent of the detector's efficiency and relies only on an accurate energy calibration.

Therefore, Monte Carlo simulations of the detector response and the subsequent unfolding¹ of the measured spectrum are not required, and the measured spectrum can be analyzed directly (though for the SPE method in section 5.3, an additional simulation of the measured spectrum is needed). The tube voltage can be determined independently of the specific spectral shape (i.e., the chosen filtration), allowing suitable additional filtration to be selected in order to improve the measurement accuracy. Specifically, additional filtration can be used to reduce the incoming count rate and minimize pile-up effects without affecting the determination of the end-point energy. Furthermore, accurate alignment of the detector towards the focus point is not mandatory (although it is recommended), contrary to “full” spectrometry, where the detector alignment is crucial.

However, accurate energy calibration is needed for the voltage determination. This can be achieved using photons emitted by radionuclide sources, because the energy of main gamma lines of radionuclides is known with very low uncertainty. Commonly used radionuclides for the detector calibration are for example Am-241, Co-57, Ba-133 and Ce-139. The energy range of the full-energy peaks should cover the energy range of interest, and only non-overlapping gamma peaks should be used for the calibration. The user needs to determine the peak centroid accurately from the measured radionuclide spectra. With CdTe spectrometers, additional effects caused by charge carrier trapping effects (CCTE) described in (Šolc et al., 2023) should be taken into account for the best accuracy. It is recommended to use a second order polynomial instead of a linear fit to obtain the energy calibration (conversion of bin ordinary number into the energy).

Nowadays in-beam X-ray spectrometry is performed with semiconductor detectors with germanium (HPGe) or cadmium telluride (CdTe) sensing elements. An example of a CdTe spectrometer is shown in Figure 5.1. It is a compact detector with the dimensions of 7 cm × 10 cm × 2.5 cm (width, length, height), excluding a collimator and a detector extender. It consists of the sensor, a preamplifier, a digital pulse processor and multichannel analyser, and a power supply. The sensor with 1 mm thickness and 3 or 5 mm square side length is mounted onto a Peltier thermoelectric cooler, reducing the sensor temperature to about 220 K to decrease thermal noise. Entrance window is made of a thin layer of a light material to minimize the absorption and scatter. The detector is suitable for X-ray spectrometry up to at least 300 keV.

¹ The computational process of reconstructing the true incident photon fluence distribution from measurements made by a detector whose response is distorted by factors such as detection efficiency, photon scattering, finite energy resolution, recombination of charge carriers, etc.



Figure 5.1: A compact X-ray spectrometer (Šolc et al., 2025a).

5.2 Linear extrapolation from measured spectrum

In the limit of perfect energy resolution, the spectrum end-point would simply be taken from the first bin from the right that has a number of counts above the background level. However, in reality, the energy resolution causes a “tail” on the right side of the spectrum and taking the first bin with higher counts than background gives an overestimation of the end-point. One method to obtain the end-point with realistic detectors is to extrapolate the end of the spectrum before the tail region using a linear fit, and the voltage is estimated from the crossing point with the x-axis (when there is no significant pile-up). The linear fit (used for the extrapolation) is calculated in the part of the spectrum as close to the edge as possible, but not in the tail region. Also, there should be a sufficient number of points for a reliable fit. The choice of the limits is subjective to some degree.

The method is visualized in Figure 5.2. The N-30 spectrum was measured with a HPGe spectrometer. In this example, the linear fit was made using the points in the energy range from 29.5 keV to 29.8 keV, and the extrapolated end-point was approximately 30.1 keV. Hence the estimated tube voltage was 30.1 kV. On the right edge, the first point below 30 keV should not be included in the fit, since it is clearly in the tail region. Whether or not to include the second to last point below 30 keV is not that clear.

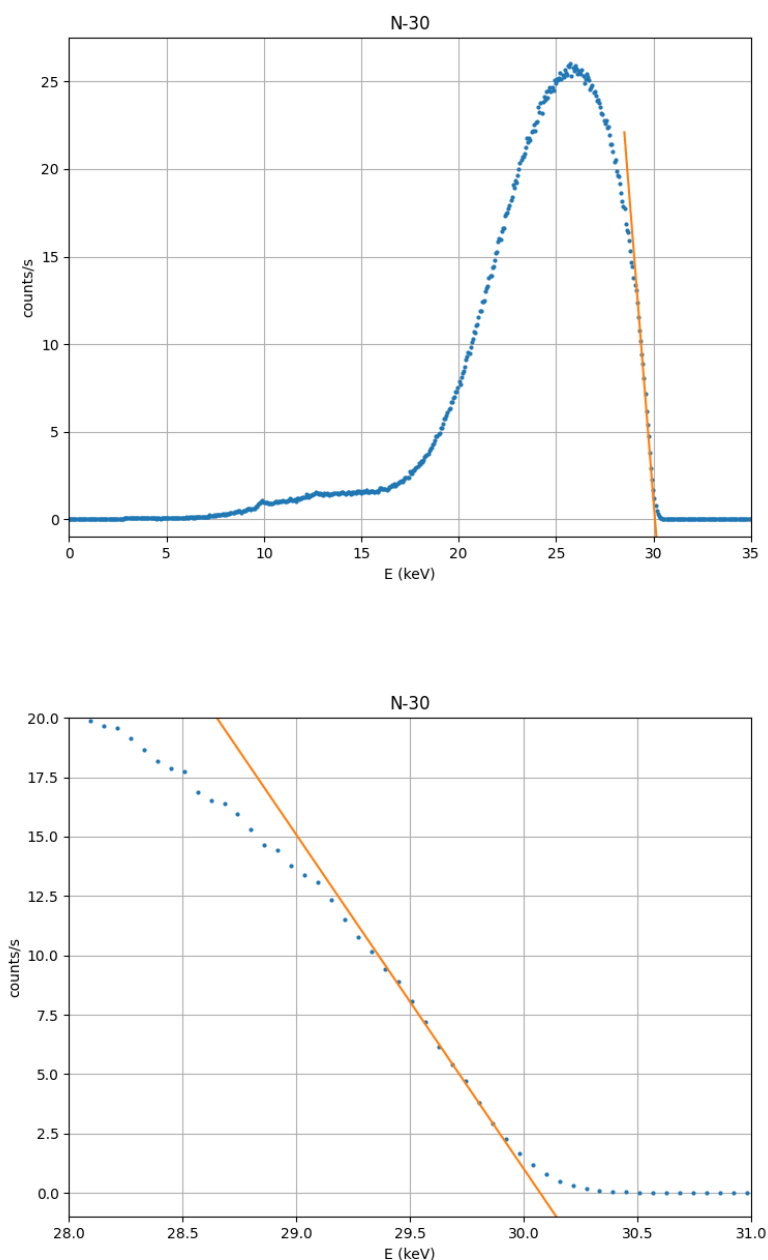


Figure 5.2: Example of the spectrum end-point extrapolation method. The extrapolated voltage for this spectrum was 30.1 keV.

It should be noted that the extrapolation method does not correspond in a theoretical sense exactly to the spectrum true end-point. It is simply a practical method to obtain an estimation of the end-point, and the method has an uncertainty. To estimate whether the method is accurate enough is up to the user, but ISO 4037 voltage measurement uncertainty limits for example for $h_K^*(10)$ can be fulfilled with

this method from N-25 onwards (ISO 4037-1:2019 Table 7, see also the next chapter and comparison with the SPE method). More information about this method can be found in (de Vries, 1995) and including the measurement uncertainty, can be found in (Rinaldi et al., 2026).

5.3 SPE method

The spectrometric practical end-point (SPE) method is a method for determining the maximum energy of the X-ray spectrum by means of X-ray spectrometry. A quantity obtained with this method is the spectrometric practical voltage (SPV) introduced by Šolc et al. (2025b). It represents the constant voltage on the X-ray tube that was obtained from the measured maximum energy of the X-ray spectrum using the SPE method. The SPV agrees with the practical peak voltage (as defined in IEC 61676 (IEC, 2023)) within 0.03 % when the voltage ripple of a high-voltage generator is below 2 %.

The SPE method is based on the comparison of the high-energy part of the measured detector spectrum (i.e. amplitude spectrum of counts in the detector) with a simulated detector spectrum obtained using Monte Carlo calculations. Because the energy of primary electrons is known in the simulations, the SPV can be found by matching the end part of the measured detector spectrum with the simulated detector spectrum. The Monte Carlo model of the detector must be available and validated. The energy resolution (and CCTE if a CdTe detector is used) must be applied on the simulated spectrum. No unfolding of the fluence spectrum is needed.

In comparison to the linear fit method, the SPE method takes into account the changes of the detector spectrum caused by CCTE in the sensor relevant for CdTe spectrometers.

The detector must be accurately calibrated. The energy calibration should consider detector energy resolution, influence of CCTE (for CdTe detectors) and should be performed at similar conditions as occurring in X-ray spectrometry (in the decreasing order of importance for a CdTe detector: sensor temperature, count rate, dead time, and collimation).

Basic steps of the SPE method:

- Acquire a detector spectrum; preferred radiation qualities are narrow photon fluence spectra, e.g. N-series radiation qualities according to ISO 4037-1
- Calculate a corresponding theoretical photon fluence spectra for radiation qualities applied in measurements; use Monte Carlo or semi-empirical tools (e.g., MCNP, EGSnrc, SpekPy) for the expected tube voltage and using at least basic information about the geometry as an input for calculations (i.e., anode material and angle, filtration material and thicknesses);
- Use the theoretical photon fluence spectrum as the particle source in Monte Carlo simulation (e.g. MCNP, EGSnrc) of the spectrometer response. Calculate the simulated detector spectrum and apply all detector corrections (i.e., CCTE, energy resolution);
- Compare the end parts of the measured and the calculated detector spectra (Figure 5.3). If there is a difference, move the calculated spectrum by a constant energy to get the best

agreement of the end part of the calculated spectrum with the measured spectrum. SPV equals to the energy of monoenergetic electrons used for the generation of the calculated fluence spectrum, corrected by the constant energy offset.

The SPV uncertainty was estimated in (Šolc et al., 2025b) to be 0.13 kV (at tube voltage 20 kV) to 0.33 kV (at tube voltage 300 kV), at $k = 1$.

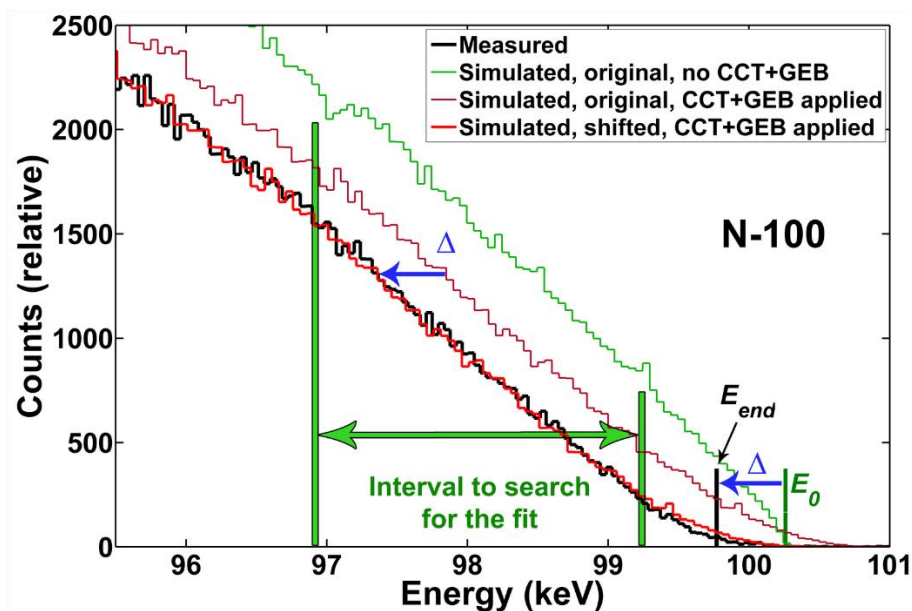


Figure 5.3: SPV determination by comparison of measured and simulated detector spectrum - the simulated detector spectrum after application of energy resolution and CCTE (dark red) was shifted by a constant energy Δ to fit the measured spectrum (black). The SPV (tube voltage) applied at the measurement is then obtained from the known energy of primary electrons in the simulation and the shift Δ (Šolc et al., 2025b).

Any laboratory that has established in-beam X-ray spectrometry using an unfolding method should already have the tools required for SPV determination using the SPE method. These include a well-characterized detector in terms of energy calibration, energy resolution, and detection efficiency, as well as its validated Monte Carlo model, tools that are otherwise required for the calculation of the detector's response matrix.

Narrow spectra radiation qualities are suitable for the SPE method, because the fluence of low energy photons, not contributing to the SPE method, is largely decreased by thick added filtration. Even more suitable and versatile is a dedicated set of filtrations put just in front of the detector, e.g. onto collimation (Figure 5.4). Advantages include independence on filtrations available at the measurement site, similarity of detector spectra between different sites reducing the number of necessary Monte Carlo simulations, and a possibility to design filters in such a way to get, for the constant tube current

and the same source to detector distance, a similar detector count rate for each combination of tube voltage and a filter dedicated to such voltage.

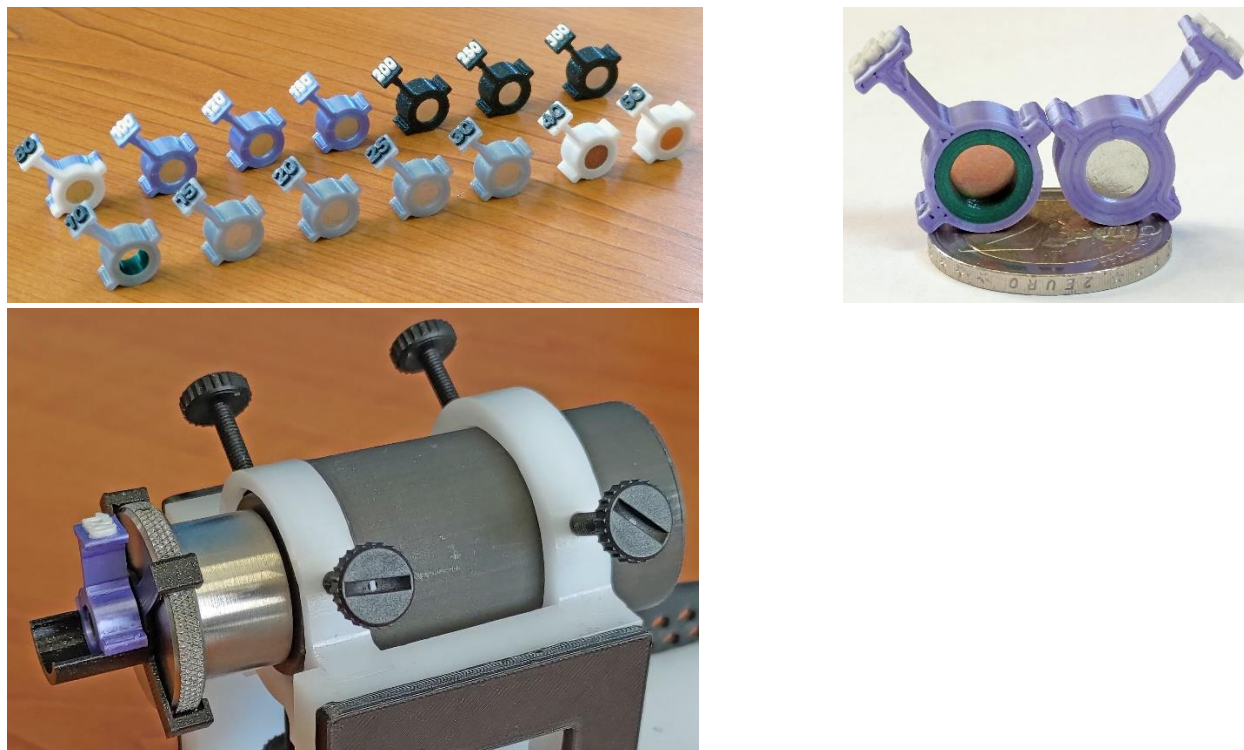


Figure 5.4: A set of filters dedicated for spectrometric measurement of tube voltage in a range from 10 to 300 kV. The filters are put onto spectrometer collimator.

6 X-ray multimeters

X-ray multimeters are devices that can calculate several different parameters related to X-ray beams, based on a single exposure. The calculations are performed based on the ratio of signals on different detectors positioned behind different filters, and complex algorithms. The principle of XMM operation is described in more details in reference (Lindstrom, 2016). One of the parameters that is commonly evaluated is the tube voltage. Different XMMs measure different quantities such as the average voltage, kVp, average peak voltage, PPV and others (Komatina et al., 2025), but as previously mentioned, ISO 4037 references ISO 16526-1, which describes the measurement of the average voltage (ISO, 2011a). Therefore, if an XMM measures average voltage, it can be used without further consideration of the measured quantity, otherwise ripple should be considered. IEC standard 61676 gives requirements for XMMs measuring PPV (IEC, 2023). Standards for measurement other tube voltage quantities are not available, but recent research shows that XMMs have similar performance when measuring other quantities (Kržanović et al., 2026).

6.1 Principle of measurement

XMM response depends on many influence quantities, most importantly radiation quality. XMMs are generally used in diagnostic radiology, and the range of radiation qualities that they usually cover includes the RQR series described in IEC 61267 (IEC, 2025), but it usually does not include N-series from ISO 4037-1 (ISO, 2019), as well as most of the other radiation qualities used in radiation protection. Although the *response* of X-ray multimeters (XMMs) depends strongly on radiation quality, the actual tube voltage generated by the X-ray system is primarily determined by the high-voltage generator and generally does not depend on the added filtration used to establish a specific radiation quality. Therefore, provided that the generator operates in a stable and reproducible manner and no special operating modes are activated for different radiation qualities, the same tube voltage setting may be assumed for different radiation qualities established at the same nominal tube voltage. For example, a measurement of tube voltage performed using RQR 6 radiation quality, which is established for 80 kV, is also valid for N-80, W-80 and H-80 radiation qualities.

RQR series is uniquely convenient for tube voltage measurements using XMMs. Radiation qualities start from RQR 2 (40 kV) and end with RQR 10 (150 kV), and 6 radiation qualities from N-series have RQR radiation qualities with corresponding tube voltages. However, laboratories using mainly ISO 4037-1 radiation qualities may need to establish the corresponding RQR radiation qualities separately in order to use XMMs under their intended measurement conditions. An important limitation of this method is that a relatively small range of tube voltages can be covered by most XMMs, e.g. between 40 kV and 150 kV. Mammography radiation qualities with tungsten anode and aluminum filtration can be used to extend this range down to 25 kV, but the more strict requirements for uncertainty in ISO 4037 for lower tube voltages are much harder to meet with XMMs than for higher tube voltages.

Measurements are performed by simply positioning an XMM in the radiation beam, performing an exposure and reading the value. For best accuracy, it is important to perform measurements in controlled conditions and for the range of influence quantities for which the XMM is designed. Accuracy can be further improved if a separate calibration of the XMM is performed in the same or similar conditions, in another calibration laboratory with traceable tube voltage measurements.

6.2 Measurement uncertainty

IEC 61676 gives limits of variation of XMM response, which can be used to calculate measurement uncertainty (IEC, 2023). The breakdown of the uncertainty components is given in table 6.1.

Table 6.1. Combined standard uncertainty for XMMs based on IEC 61676 limits of variation

Influence quantity	IEC 61674	
	Limits of Variation [%]	Relative standard uncertainty [%]
Voltage waveform and frequency	2	1.15
Anode angle	0.5	0.289
Filtration	1.5	0.866
Dose rate	0.5	0.289
Irradiation time	0.5	0.289
Field size: Rated range	0.5	0.289
Field size: large	2	1.15
Detector-Focal distance	0.5	0.289
Angle of incidence	0.5	0.289
Rotation	0.5	0.289
Temperature and relative humidity	1.0	0.577
Power supply	0.5	0.289
Electromagnetic compatibility	1.0	0.577
Additional tungsten filtration (tube aging)	2.0	1.15
Combined standard uncertainty (calibration factor is omitted from the table), $k = 2$		4.9

The table shows that even without additional calibration, XMMs can provide measurements with an uncertainty of around 5 %. This level of measurement uncertainty is suitable for establishing matched radiation fields between N-60 and N-150. In a recent study, 25 XMMs from 6 different manufacturers were tested, and the XMMs generally satisfied the standard limits of variation. Even though IEC 61676 uses PPV, the same limits were applied for other tube voltage quantities (Kržanović et al., 2026).

It is possible to reduce the uncertainty by performing dedicated calibrations, where the conditions during the calibration and during the use of the XMM are matched as closely as possible. Assuming a good match between anode angle, filtration, irradiation time, detector-focal distance, and possibly also field size and dose rate, uncertainty can be significantly reduced. In a recent comparison, measurement uncertainties between 2.8 % and 3.5 % ($k = 2$) were reported (Stojanovic et al., 2026a; Stojanovic et al., 2026b).

7 Comparison between different tube voltage measurement methods

7.1 Multilateral on-site comparison

A comparison of tube voltage measurements was performed in Institut za mjeriteljstvo Bosne i Hercegovine (IMBiH) SSDL, with the participation of Czech Metrology Institute (CMI), Vinča Institute of Nuclear Sciences - National Institute of the Republic of Serbia, University of Belgrade (VINS) and IMBiH. The comparison was performed for 9 tube voltage values, using a voltage divider calibrated against the SPE method, two spectrometry methods (SPE and linear extrapolation), calibrated XMMs and uncalibrated XMMs. The comparison has shown that all the mentioned methods can be used to produce results with measurement uncertainties sufficient for ISO 4037-1 requirements, at least in the higher part of the tube voltage range. Comparison reference values were calculated as the weighted mean of two spectrometry methods. In total, 89 out of 91 results were in agreement with the comparison reference value (*CRV*), within the measurement uncertainty (Stojanovic et al., 2026a; Stojanovic et al., 2026b).

The measurement uncertainties were in the interval between 0.3 % and 0.75 % for SPE, 1.1 % and 1.4 % for linear extrapolation method, 2.8 % and 3.6 % for calibrated XMMs, and 4.9 % for uncalibrated XMMs (all with $k = 2$). Calculation for the last case is also shown in table 6.1, and further explanation can be found in (IEC, 2023).

7.2 Spectrometric methods

The comparison of SPV with the tube voltage determined with the linear fit method is presented in Figure 7.1. The results are valid for tube voltage determined using narrow spectra and for spectrometry using a CdTe detector (X-123CdTe, Amptek). The results can be separated into two groups - up to 67.5 kV and above 67.5 kV, and fitted with a straight line. The fits are as follows:

$$\text{For } U \leq 67.5 \text{ kV:} \quad \Delta U = -1.256 \times 10^{-1} + 1.451 \times 10^{-3} \times U$$

$$\text{For } U > 67.5 \text{ kV:} \quad \Delta U = -5.365 \times 10^{-1} + 7.535 \times 10^{-2} \times U$$

where U is the tube voltage in kV and ΔU is the difference (in kV) between SPV and the linear fit ($SPV - U_{\text{linear}}$).

As a result, if a laboratory uses a X-123CdTe detector for spectrometric determination of tube voltage, the laboratory may choose to use a simple method of linear fitting of the measured spectrum instead of a more complicated SPE method, and then apply the above-mentioned correction to account for the influence of CCTE in CdTe sensor. Otherwise, the difference should be considered in the uncertainty budget.

The equations assume that the energy calibration valid for unfolding (a shift of full-energy peak maximum due to CCTE is considered) was applied on the spectrum used for linear fit. Otherwise, if a

simple energy calibration for visualisation was used (peak maximum is the gamma photon energy), a difference between energy calibration for unfolding and for visualisation should be included, reaching up to about 0.2 keV (see (Šolc et al., 2025b) and Figure 11 therein).

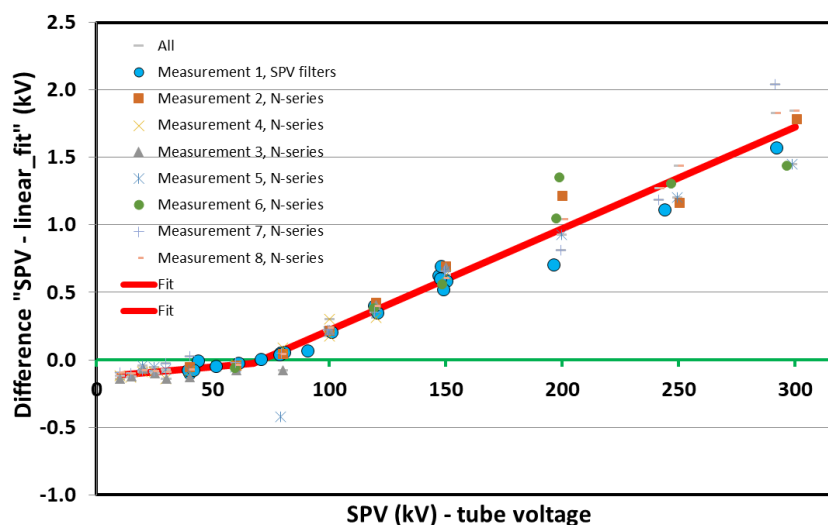


Figure 7.1: The difference between the tube voltage determined with the SPE method and with a linear fit of the end of detector spectrum, as a function of nominal tube voltage. Measurements were performed at various sites across Europe.

7.3 SPE method to voltage divider

Examples of comparison of SPV to voltage divider readings are presented in Figures 7.2 and 7.3. In Figure 7.2, a comparison of SPV to calibrated voltage divider performed at X-ray beams obtained using a 450 kV high voltage generator at Physikalisch-Technische Bundesanstalt (PTB), Germany, was performed (Figure 7.2; Šolc J. et al., 2025b). Obtained SPV values were lower by roughly 0.2 kV for the tube voltage up to 150 kV, then the differences slightly increased up to 1.2 kV at a tube voltage of 300 kV. Up to 200 kV the differences fitted within the measurement uncertainty at $k = 2$.

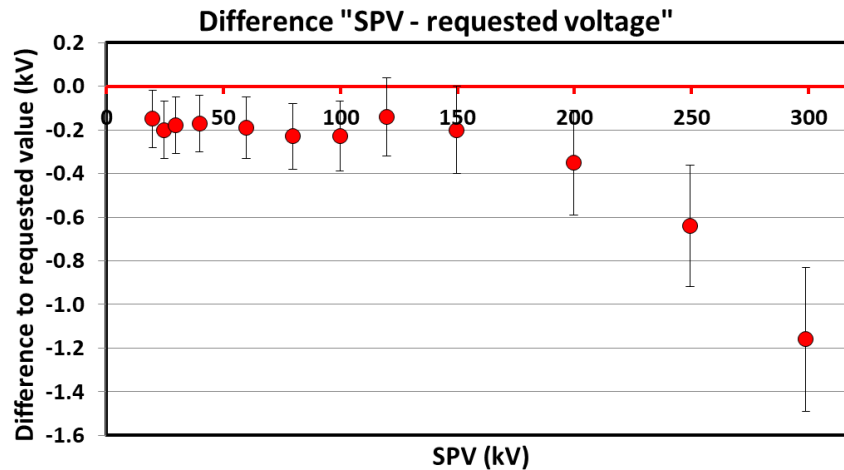


Figure 7.2: The difference between the SPV and the voltage divider reading at PTB, Germany. The error bars show uncertainties at $k = 1$ (Šolc J. et al., 2025b).

Yellow triangles and green diamonds of Figure 7.3 show the comparison of SPV with nominal values of tube voltage calibrated against a voltage divider and obtained at X-ray beams of industrial X-ray generator ISOVOLT TITAN E 160 kV. The measurements were performed at Sateilyturvakeskus (STUK), Vantaa, Finland, in the interval of tube voltages from 10 kV to 150 kV. The SPV was determined from N-series and RQR-series spectra. Excellent agreement was obtained with nominal voltage values.

For comparison, SPV measurements at X-ray beams generated by a 320 kV high voltage generator at the same irradiation room are presented in the same Figure 7.3 (red circles). The nominal values were not calibrated against a voltage divider. For this 320 kV generator, the measured SPV was lower by 0.4 kV - 0.7 kV than the nominal tube voltage.

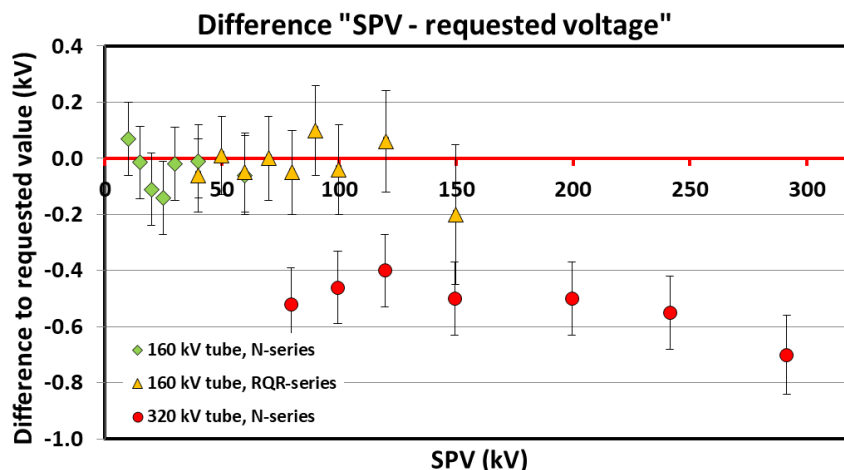


Figure 7.3: The difference between the SPV and the expected tube voltage on industrial X-ray generators ISOVOLT TITAN E 160 kV (diamonds - N-series qualities, triangles - RQR-series qualities) and 320 kV (circles - N-series qualities).

7.4 SPE method to nominal tube voltage

Figure 7.4 shows the results of SPV measurements at clinical X-ray equipment. The measurements were performed at Städtisches Klinikum in Braunschweig (SKBS), Germany, on the stationary interventional C-arm X-ray imaging system Artis zee multi-purpose (Siemens Healthineers) with MEGALIX Cat Plus 125/20/40/80-GW X-ray tube. The SPV was compared to the nominal tube voltage ranging from 50 kV to 125 kV set by the operator. At each voltage, spectra were acquired for several additional Cu filtrations. Excellent agreement within 0.1 kV was obtained. Also, the results presented here and in Figure 7.3 demonstrate that the spectrum end-point energy determined with the SPE method does not depend on the spectrum filtration.

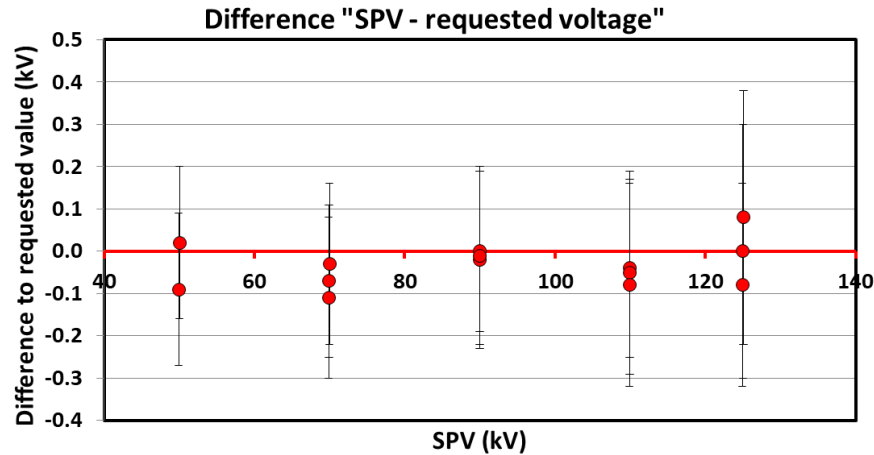


Figure 7.4. The difference between the SPV and the nominal tube voltage set by operator on stationary interventional C-arm X-ray imaging system Artis zee multi-purpose (Siemens Healthineers) at SKBS, Braunschweig, Germany.

Figure 7.5 summarizes SPV measurements at CMI, Prague, Czech Republic, where the tube voltage has not been determined previously and the tube voltage was taken only from the value set on the display panel. The measurements in X-ray beams generated with 160 kV high voltage generator (ISOVOLT 160 HS and an MB 161/4 X-ray tube) obtained SPV values, which were constantly higher roughly by 0.8 kV than the set tube voltage value, in the voltage range from 10 kV to 150 kV. The differences up to 3.2 kV were obtained for nominal voltages from 200 kV to 300 kV realized with a 320 kV high voltage generator (PANTAK PMC with an MB 350/1 X-ray tube; Šolc J. et al., 2025b). These measurements allowed to determine the new values to be set on the control panel in order to obtain the correct tube voltage required by relevant standards.

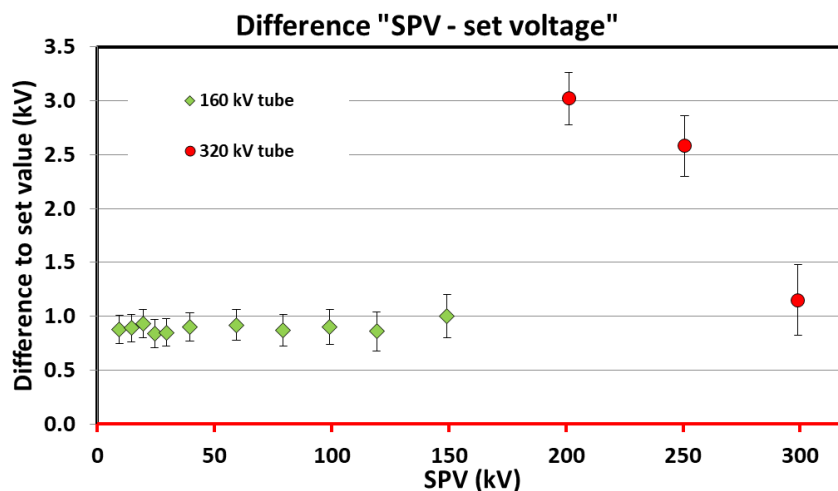


Figure 7.5: The difference between the SPV and the expected tube voltage, (set), set on the 160 kV (diamonds) and the 320 kV (circles) X-ray generator at CMI. The error bars show uncertainties at $k = 1$.

7.5 TraMeXI comparison

A supplementary comparison was organized within 22NRM01 TraMeXI project, a normative project within the European Partnership on Metrology. The comparison is registered in the Key Comparison Database as EURAMET.RI(I)-S20 comparison (BIPM, 2026b). The comparison results are not available at the moment of the publication of this guide.

8 Conclusion

While the invasive tube voltage measurement by voltage divider can be the preferred method with respect to the accuracy of the measurement that it can deliver, tube voltage measurements by spectrometry or non-invasive measurement devices also allow achieving uncertainties that are suitable for ISO 4037-1 requirements at least for a limited voltage range. All methods have their advantages and disadvantages.

Voltage dividers provide the smallest uncertainty, allow measuring the ripple, and can be used for any tube voltage and any radiation quality. They also allow real-time monitoring of tube voltage. On the other side, they are expensive and complicated to calibrate (unless a local calibration laboratory is available), and they introduce additional components in the X-ray electrical circuit such as additional high voltage cables, which may increase the possibility for problems. Furthermore, voltage dividers measure generating potential, which may be different from tube potential in those cases where a protective resistor is installed in the x-ray tube housing, making them not useful to determine the end-point energy of the photon spectrum without further measures.

Spectrometry can be used for any tube voltage and any radiation quality. This methodology allows achieving an intermediate uncertainty when compared to voltage dividers and XMMs. The tube voltage is directly estimated measuring the spectrum end-point. Although the spectrometer setup and characterization are relatively complex (compared to XMMs), the methodology to estimate the tube voltage from the measured photon fluence spectrum is comparatively straightforward as it does not require spectrum unfolding and simple methods are available. Commercially available spectrometers are increasingly affordable. However, radionuclide sources are required for energy calibration. Spectrometry does not provide information about waveform, nor does it offer real-time information during the regular X-ray use. Nevertheless, once this method is established at a given laboratory, monitor chamber can be used to monitor beam stability.

XMMs are very simple to use and relatively cheap. The uncertainty of measurements is relatively large when compared with the above two methods. The uncertainty can be reduced by performing dedicated calibrations, but it is expected to still be larger than for the other two methods. Another disadvantage of this method is the measurement range of XMMs which is relatively small, e.g. between 40 kV and 150 kV. XMMs do not provide real-time information during the regular X-ray use, but as for the case of the spectrometric method, a monitor chamber can be used to monitor beam stability.

A summary of the methods is given in Table 8.1.

Table 8.1: Summary of methods for determination of high voltage of X-ray tubes

	Voltage divider	Spectrometry - SPE method	Spectrometry - linear method	XMM
Type	Invasive	Non-invasive	Non-invasive	Non-invasive
Range (kV)	Covers the whole range of X-ray tube voltages	Within the range of detector energy calibration; feasible from 5 kV to at least 350 kV	Within the range of detector energy calibration; feasible from 5 kV to at least 350 kV	Depending on the model, but usually not lower than 25 kV and not higher than 150 kV
Typical measurement uncertainty ($k = 2$)	0.1 % - 2 %	0.35 % - 0.75 %	1.3 % - 3 %	2.5 % - 5 %
Ripple sensitivity	Low sensitivity, voltage dividers can record time resolved data	None up to about 4 % ripple	Low sensitivity up to about 4 % ripple	Sensitive to ripple; some models can record time resolved data
Real-time capability	Yes	No	No	No

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